

Salience-Guided Question Generation: Feature Ablations, Methodological Insights, and Issue Analysis

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Agenda

- Problem & Motivation
- Proposed Approach & Features
- Experimental Setup & Results
- Qualitative Analysis & Insights
- Limitations & Future Work

Problem & Motivation

- Standard question generation is often fluent but poorly grounded – questions may ignore key evidence in the passage.
- **SQuAD benchmark:** 100,000+ QA pairs, 3–5 questions per passage, answers are exact extractive spans.
- Challenge: How to steer the LLM toward the most answer-bearing sentences?
- Goal: Improve grounding and answer consistency using sentence-level salience guidance.

Evaluated on 750 SQuAD passages.

Salience-Guided QG Pipeline

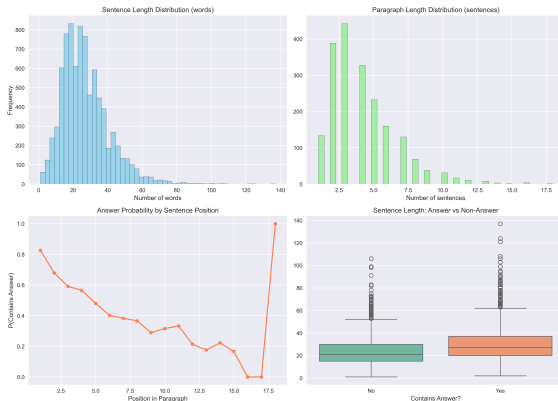
1. Score every sentence using linguistic + surprisal features.
2. Select top-k most salient (answer-bearing) sentences.
3. Generate questions from only the selected salient evidence using Qwen2.5-7B-Instruct.
4. Evaluate with overlap metrics + QA consistency.

Core Design Principle

Soft grounding via salience guidance – prioritize important sentences while keeping questions fluent and diverse.

Sentence Raw Data: Visual Snapshot

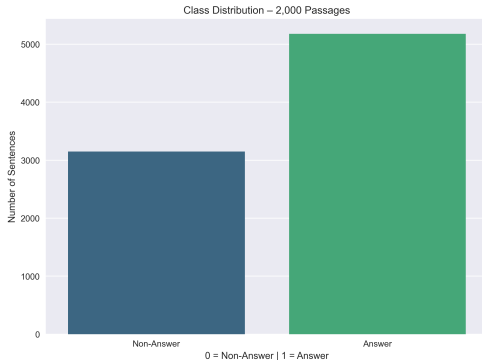
What the raw sentence data looks like before modeling



Raw sentences are short on average, paragraph lengths vary, and answer-bearing sentences tend to appear earlier and be slightly longer.

Class Imbalance: Skewness in 2,000 Passages

Answer-bearing sentences are more frequent than non-answer sentences in the 2,000-passage set



Takeaway: The data is imbalanced, so precision/recall trade-offs matter when evaluating the salience classifier.

Feature Family	Rationale
Linguistic (length, position, POS ratios, lexical density, discourse markers)	Interpretable, fast, strong structural priors for important sentences.
GPT-2 surprisal stats	Captures local uncertainty and information density.
BERT surprisal stats	Captures contextual uncertainty with bidirectional context.

Linguistic Features (14 total)

Category	Features
Surface	sentence_length_words, sentence_position, sentence_position_norm, avg_word_length
Lexical	type_token_ratio, lexical_density
POS ratios	noun_ratio, verb_ratio, adj_ratio, pronoun_ratio, noun_verb_ratio
Discourse	named_entity_density, causal_marker_ratio, contrast_marker_ratio

Experimental Setup (750 Passages)

Three salience conditions (same prompt, same LLM):

- **Full Salience:** 26 features (linguistic + surprisal)
- **Linguistic Only:** 14 linguistic features
- **Top-10:** Compact feature set
- **Baseline:** No salience guidance (random sentences)

Same evaluation pipeline for all conditions.

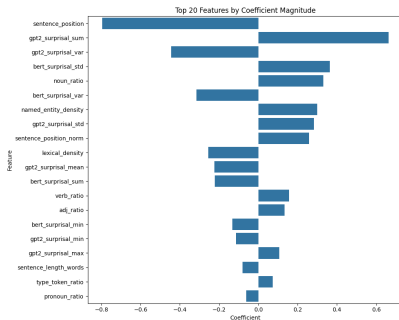
Salience Classifier: Classification Metrics (2000 Passages)

Configuration	Features	Accuracy	Precision	Recall	F1-Score
Full (Ling + Surprisal)	26	0.693	0.786	0.695	0.738
No Surprisal (Ling only)	14	0.692	0.776	0.710	0.742
Top-10 Compact	10	0.674	0.765	0.687	0.724

Quick Summary: Linguistic features are strong; surprisal mainly boosts precision.

Feature Importance: What Drives Saliency?

Top-ranked features from the 2000-passage saliency classifier



Quick Summary: The ranking shows that simple linguistic cues remain highly informative, while surprisal features add complementary signal for saliency prediction.

Question Generation Results (750 Passages)

Salience vs Baseline – 5 Key Metrics

Generated with Qwen2.5-7B-Instruct | Evaluated with deepset/roberta-base-squad2 (QA) & roberta-base (BERTScore)

Metric	Salience			Baseline		
	Full	NoS	Top10	Full	NoS	Top10
ROUGE-L	0.349	0.360	0.363	0.350	0.369	0.371
QA-EM	0.537	0.513	0.516	0.569	0.578	0.561
QA-F1	0.775	0.764	0.772	0.767	0.778	0.765
BERTScore	0.905	0.908	0.908	0.906	0.909	0.910
Answer-Recovery-F1	0.402	0.430	0.425	0.415	0.454	0.447

Key Observation: Baseline wins on surface overlap. Salience wins on answer consistency.

Salience vs. Baseline Metric Comparison

Baseline Strength

- Better lexical overlap & BERTScore
- Closer surface match to gold questions

Salience Strength

- Better QA consistency & answer recovery
- Stronger grounding to important evidence

Takeaway: Grounding is not the same as gold overlap.

Why Questions Get Flagged

Main reasons (from 2302 issues in qa_generation_issues.json)

- Low lexical overlap with gold question
- QA model answer does not match generated answer
- Generated question targets a different (but valid) fact
- QA model answer weakly aligned with gold answers

Key Insight: Many flagged questions are not low-quality – they simply focus on different valid facts from the salient sentences.

Data-Sourced Example 1: Better Semantic Alignment

Warsaw_Pact (example_index 3) — Seeking Specificity

Gold (SQuAD question): What did the Soviets declare about the North Atlantic Pact?

Baseline

Question: Why did the Soviets propose a new agreement?

Answer: to accept the participation of the USA

QA consistency F1: **0.67**

Salience (Ours)

Question: What did the Soviets declare about the NATO?

Answer: readiness to examine jointly with other parties

QA consistency F1: **0.94**

Salient sentence: “The Soviets declared their readiness to examine jointly...”

Insight: Salience grounds on more specific treaty language, improving QA consistency by 40%. Source: qa_generation_issues.json (example_index 3)

Data-Sourced Example 2: Grounding Recovers Details

Warsaw_Pact (example_index 10) — Treaty Framework

Gold (SQuAD question): How many countries formed the initial Warsaw Pact membership?

Baseline

Question: What did Warsaw Pact members pledge?

Answer: mutual defense

QA consistency F1: **0.36**

Saliency (Ours)

Question: What were treaty relations based on?

Answer: mutual non-intervention, sovereignty, independence

QA consistency F1: **0.70**

Salient sentence: "Relations among treaty signatories were based on mutual non-intervention..."

Insight: Saliency captures nuanced treaty principles; QA consistency improves by 95% via better passage grounding. Source: qa_generation_issues.json (example_index 10)

Issue-Present Case: Both Methods Flagged

Pub (example_index 414)

Gold (SQuAD question): In what historical period was a large portion of the population illiterate?

Salient sentence: "For this reason there was often no reason to write the establishment's name on the sign..."

Baseline

Question: Why were pictures useful on signs?

Answer: pictures on a sign were more useful than words

QA consistency F1: **0.0**

Salience (Ours)

Question: Why was writing the establishment's name on the sign uncommon?

Answer: pictures on a sign were more useful

QA consistency F1: **0.0**

Takeaway: This is a true issue case: both methods ask reasonable questions, but they drift from the gold fact target, causing QA mismatch. Source: qa_generation_issues.json (example_index 414)

1. Full features give best consistency; compact features give better overlap.
2. Linguistic features alone are surprisingly powerful.
3. Grounding $\neq$ gold overlap – salience targets important content.

Next: Discourse-Level Salience with RST

Current method is sentence-level only. Integrating Rhetorical Structure Theory (RST) will allow us to:

- Identify nuclear (central) vs satellite (supporting) sentences
- Capture discourse relations (evidence, elaboration, contrast)
- Model multi-sentence evidence chains

Expected impact: significantly better salience ranking and consistency on complex questions.

Conclusion & Open Questions

- Saliency-guided QG is useful when grounding matters.
- Full features improve consistency; compact features improve overlap.
- Next step: optimize for the target objective, not one metric.

Thank you.

Questions, feedback, and suggestions for the next iteration are very welcome.